Enhancing Emotional Support through Conversational AI via Big Five Personality Detection and Behavior Regulation Based on the Zurich Model

Masters Thesis

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Abstract

Objective. This thesis investigates whether dynamically regulating chatbot behavior based on user personality traits improves the effectiveness of emotional support interactions. Grounded in the Big Five (OCEAN) personality framework and the Zurich Model of Social Motivation, the study implements a personality-adaptive chatbot system that detects user traits in real time and adjusts its conversational tone, structure, and engagement strategy accordingly.

Experimental Design. Two extreme personality profiles—high OCEAN (Type A) and low OCEAN (Type B)—were simulated to evaluate system performance under contrasting psychological conditions. A controlled experiment compared personality-adaptive chatbots (regulated) with non-adaptive chatbots (baseline) across 10 simulated conversations. Regulated chatbots continuously updated their personality detection after each user message and dynamically applied behavior prompts aligned with detected traits.

Evaluation Method. Conversation quality was assessed using a structured Evaluation Matrix scored by a custom Evaluator GPT system . Each assistant's response was rated on detection accuracy, regulation effectiveness, emotional tone, relevance, and satisfaction of personality-specific emotional needs.

Results. Regulated assistants consistently outperformed baseline counterparts across all criteria and both personality types, achieving a 34.44% average performance improvement. Qualitative analysis confirmed that regulated assistants generated more empathetic and contextually aligned responses.

Conclusion. The findings support the hypothesis that real-time personality-aware regulation significantly enhances chatbot interaction quality, particularly in emotionally supportive contexts. This work demonstrates the feasibility and value of integrating personality detection with adaptive behavioral modulation in conversational AI and offers a scalable framework for developing emotionally intelligent systems for elder care, mental health, and digital companionship.

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1. Introduction

1.1 Problem Context

Psychosocial Risks of Loneliness in Aging Populations. Social isolation and loneliness are increasingly prevalent among older adults, with profound effects on emotional wellbeing, cognitive health, and overall quality of life. Studies indicate that over half of elderly individuals report persistent feelings of loneliness (Musich et al., 2015; MacLeod et al., 2018), a condition associated with increased risks of depression, cognitive impairment, and mortality (Couture, 2012; Luo et al., 2012). Structural factors such as single-person households, chronic illness, and mobility limitations further compound vulnerability, especially in aging societies like Switzerland, where up to one-third of elderly residents report social disconnection (Hämmig, 2019).

Limitations of Traditional Support Systems. Conventional interventions, community programs, in-person therapy, or family support—are resource-intensive and not consistently accessible. Technological alternatives, such as digital platforms and AI companions, offer scalable, continuous interaction capabilities, especially for those facing mobility constraints or living alone (Anttila et al., 2020; Marottoli et al., 2007). However, effectiveness varies based on how well such systems recognize and respond to users' nuanced emotional and psychological needs.

Emergence of Personality-Aware Conversational AI. Advances in large language models (LLMs) have enabled conversational AI systems to simulate social interaction. Tools like

Replika and ElliQ provide companionship and emotional support through dialogic engagement (Ta et al., 2020; Broadbent et al., 2024). Evidence suggests that users form affective bonds with such systems, particularly when interactions are perceived as emotionally attuned (Xie & Pentina, 2022; De Freitas et al., 2024). However, existing systems lack consistent frameworks for dynamically tailoring their responses to diverse user personalities.

Need for Personalized Emotional Regulation. Affective computing now supports not just emotion detection but real-time modulation of conversational behavior. Yet, many systems still operate on generic prompts or static user profiles, limiting emotional alignment and responsiveness. This study responds to that gap by introducing a dynamic regulation system that adjusts chatbot tone, content, and engagement strategy based on the user's evolving personality profile.

1.2 Theoretical Framework and Technological Foundations

Big Five Personality Trait Model. The Five-Factor Model (OCEAN: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) is a foundational framework in personality psychology (McCrae & John, 1992). It enables structured profiling of user disposition across emotional and behavioral dimensions, offering a logical basis for adaptive interaction design.

Zurich Model of Social Motivation. To guide behavior regulation, the thesis employs the Zurich Model (Quirin et al., 2023), which conceptualizes human behavior through motivational systems related to security, arousal, and affiliation. These domains are mapped to

specific OCEAN traits (e.g., Neuroticism to security needs), forming the psychological foundation for generating personality-aligned prompts.

Large Language Models and Multimodal Extensions. The growing integration of LLMs into affective computing has produced architectures capable of empathy simulation, emotional state inference, and dynamic response generation (Zheng et al., 2023; Abbasian et al., 2023). This thesis builds on that progression by implementing a modular detection-regulation pipeline for OCEAN-based behavioral adaptation in conversational AI.

1.3 Practical Significance

Emotional Support Delivery via Chatbots. AI companions are increasingly positioned to deliver emotional support across healthcare, eldercare, and counseling domains. Yet, without adaptive mechanisms that account for individual personality differences, user engagement remains inconsistent. This study presents a structured evaluation of whether regulated chatbot responses—generated dynamically based on detected personality traits—can significantly improve perceived support quality.

Scalability and Automation Potential. The system architecture (Figure A.1.) developed in this research emphasizes modularity, traceability, and psychological coherence. By leveraging real-time detection and structured behavior mapping, the proposed system offers a scalable framework for developing socially intelligent agents applicable to various vulnerable populations, including older adults.

2. Background and Literature Review

2.1 Project Scope

Objective of the Study. This thesis investigates the effectiveness of personality-adaptive behavior regulation in enhancing the emotional support capabilities of AI assistants. It extends beyond preliminary feasibility testing to implement and evaluate a real-time detection-regulation loop using the Big Five personality framework.

System Implementation Focus. The study focuses on building an adaptive chatbot system that infers the user's personality dynamically across conversation turns and adjusts its tone and behavioral style accordingly. It simulates interactions with extreme personality types (high-OCEAN and low-OCEAN) and evaluates system performance using a structured Evaluation Matrix scored by a custom Evaluator GPT.

Application Context. The assistant is targeted at older adults at risk of social isolation, aiming to deliver scalable emotional support through psychologically grounded, dynamically modulated dialogue. The experiment's design allows for precise control over interaction variables, enabling traceable evaluation of personalization effects.

2.2 Research Objectives

Primary Objective. To evaluate whether real-time personality-adaptive regulation improves conversational quality, emotional alignment, and psychological attunement in chatbot-user interactions compared to static, non-adaptive baselines.

Secondary Objectives.

- To implement a cumulative personality detection module that updates the user's OCEAN vector throughout the conversation.

- To apply behavior regulation via Zurich Model–aligned prompts mapped to detected traits.

- To compare regulated and baseline chatbot responses using multidimensional evaluation criteria.

- To identify strengths, limitations, and scalability potential of personality-aware conversational AI in support contexts.

2.3 Key Concepts and Definitions

Personality-Aware Emotional Support AI. A conversational agent that dynamically adapts its behavior in alignment with the user's personality profile, with the goal of enhancing psychological safety, emotional resonance, and perceived empathy.

Big Five Personality Traits. A trait-based psychological framework defining five major personality dimensions—Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism—that inform individual differences in social interaction.

Behavior Regulation. The process by which AI-generated responses are modified in real time based on user personality traits. This includes tone, topic framing, emotional intensity, and engagement strategy.

Zurich Model–Aligned Prompting. A behavior design mechanism that links personality traits to core motivational systems (security, arousal, affiliation), guiding the chatbot's regulatory actions.

Evaluation Matrix. A structured format for scoring chatbot responses on five regulated and three baseline dimensions, including detection accuracy, regulation effectiveness, emotional tone, coherence, and psychological need fulfillment.

2.4 Research Scope and Delimitations

Experimental Boundaries. The study focuses on short-form text-based interactions (six turns per dialogue) with two simulated user profiles representing the extreme ends of the Big Five personality spectrum. It does not attempt to generalize findings across live or longitudinal user populations.

Technical Constraints. Interactions were conducted using a controlled system architecture with fixed trait detectors and curated prompts. No voice, facial, or physiological data were used; personality detection was performed solely through language-based input. Evaluation was limited to textual content and was conducted via an LLM-based Evaluator OpenAI GPT without human raters.

Exclusions. The study does not explore cross-cultural differences in personality expression, user demographic variation beyond age category, or hybrid multimodal interaction contexts. Additionally, ethical considerations such as user consent, data security, or emotional dependency risk were acknowledged but not directly studied in this implementation phase.

2.5 Literature Review and Current State of Research

Challenges and Evolution in Affective Computing. Current AI models face significant challenges in emotional recognition and personalization. Poor generalization across individuals, contexts, cultures, and languages remains a notable limitation (Alisamir & Ringeval, 2021). Algorithms often struggle with subtle or complex emotions, particularly in nuanced human interactions (Qasim, 2022). The field of affective computing has evolved beyond simple emotion recognition towards simulation and stimulation, introducing new technical complexities and ethical considerations, such as the need for interactional respect (Alberts et al., 2024).

Advancements with LLMs and Multimodal Systems. Recent progress in emotional support assistants is largely driven by Large Language Models (LLMs) and multimodal capabilities. Methodologies for building extensive emotional support dialogue datasets are emerging (Zheng et al., 2023), alongside cascaded neural networks integrating LLMs with dialogue state recognition to enhance emotional comprehension (Chen et al., 2023). Multimodal systems like EmoAda combine various data streams (video, audio, text) to offer scalable psychological support (Dong et al., 2024). Other innovations include Conversational Health Agents (CHAs) interpreting multimodal cues for context-aware responses (Abbasian et al., 2023), Brain-Computer Interfaces (BCIs) like ARIEL monitoring emotional states for adaptive support (Sorino et al., 2024), and physiology-driven LLMs using wearable data for stress prediction and empathy (Dongre, 2024).

AI Companions and Digital Interventions for Loneliness. Specific applications like AI companions (e.g., Replika, ElliQ) show potential in elder care, offering companionship and

reducing loneliness (Ta et al., 2020; Broadbent et al., 2024). However, many rely on fixed strategies lacking dynamic adaptation. Similarly, digital technology interventions (DTIs) for loneliness show mixed results (Shah et al., 2019; Shah et al., 2020), often limited by a lack of evolving personalization, which hinders sustained engagement.

Personality-Aware Dialogue Systems. Incorporating user personality is a growing area. Frameworks like PROMISE (Wu et al., 2023) and personality-aligned prompting demonstrate improved coherence. However, many systems use static, one-time personality initialization. This thesis advances the field by implementing a continuous, real-time detection-regulation loop adapting to evolving personality cues.

Regulation Models and Evaluation. While systems like EmoAda and evaluation frameworks like FEEL (Zhang et al., 2024) advance emotion tracking and assessment, few explicitly operationalize psychological regulation theories. This study addresses this by aligning behavior regulation directly with the Zurich Model of Social Motivation (Bischof, 1993), enabling trait-specific adjustments across motivational domains (security, affiliation, arousal). While frameworks like FEEL offer improvements, this study introduces a novel Evaluation Matrix scored by a custom Evaluator GPT, enabling traceable, turn-by-turn assessment of specific criteria like regulatory alignment and personality need satisfaction.

Identified Gaps and Thesis Contribution. Despite progress, the literature reveals gaps in systematic implementations combining real-time personality detection with behavior modulation grounded in established psychological theory. Few studies rigorously examine adaptive effectiveness in short-turn, high-contrast scenarios. Challenges also remain in achieving

nuanced emotional understanding and aligning LLMs for complex support tasks (Zheng et al., 2023). This work addresses these gaps through its modular architecture, simulation of personality extremes, use of the Zurich Model for regulation, and an empirically grounded evaluation methodology, contributing to the development of more personalized and effective emotional support AI.

3. Research Questions

3.1 Primary Research Question

How can an AI assistant dynamically adapt its conversational behavior to user personality traits using real-time detection and behavior regulation to improve emotional support outcomes for older users?

3.2 Sub-Questions

Personality Detection Design: How can a modular detection architecture infer the user's personality vector (OCEAN) incrementally and reliably over short-form dialogue?

Behavior Regulation Mechanism: How can Zurich Model principles be operationalized to guide tone and behavioral adjustments that reflect user-specific motivational needs?

Evaluation Impact: How does real-time behavior adaptation affect conversational quality, emotional appropriateness, and user-aligned engagement compared to static baselines?

4. Methodology

4.1 Overview

Experimental Design. The study employs a controlled experimental design comprising two primary chatbot conditions. The Regulated Assistant dynamically adapts its responses in real-time based on detected (Appendix B.4) user personality traits using personality-specific regulation prompts (Appendix B.1). The Baseline Assistant (Appendix B.2) delivers static, non-adaptive responses without considering user personality.

Personality Profiles. To effectively assess adaptive chatbot interaction capabilities, two extreme personality profiles (Appendix B.3) were simulated from the Big Five model. Personality Type A (Positive Extreme) exhibits maximum positive trait levels, while Personality Type B (Negative Extreme) reflects maximum negative trait levels. Chatbots were specifically configured as OpenAI GPTs (Appendix B.5, B.6) to interact as these clearly defined profiles.

Evaluation Framework. A systematic approach to conversation assessment was adopted using an Evaluation Matrix (Appendix B.7). This matrix measures detection accuracy, regulation effectiveness, appropriateness of emotional tone, relevance and coherence of responses, and effectiveness in addressing personality-specific emotional needs.

Evaluator GPT System. Evaluations were conducted utilizing a structured Evaluator GPT (Appendix B.8, B.9, B.10) designed explicitly for unbiased, incremental, and traceable assessments of conversation quality. This system ensured rigorous comparative analysis between adaptive and non-adaptive chatbot responses.

System Architecture. A detailed end-to-end architecture, illustrating the processes involved in both adaptive (regulated) and baseline chatbot interactions, evaluation procedures, and supporting modules, is provided in Appendix A (Figure A.1). The diagram clarifies the progression of user personality prompts through detection, regulation, and evaluation stages.

4.2 Experimental Setup

4.2.1 Personality Simulation Design

Personality Extremes Definition. Two extreme personality profiles were defined using the Big Five (OCEAN) model. Personality Type A (Positive Extreme) corresponds to OCEAN values (+1, +1, +1, +1, +1), representing users who are open, conscientious, extraverted, agreeable, and emotionally stable. Personality Type B (Negative Extreme) corresponds to OCEAN values (–1, –1, –1, –1, –1), representing users who are closed to experience, disorganized, introverted, disagreeable, and highly neurotic.

Prompt Encoding. Each personality profile was systematically encoded into specific prompts to simulate human conversational behavior. These prompts were carefully constructed to maintain internal consistency across conversation turns, aiming at realistic portrayals of expected behavioral responses for each extreme personality type.

4.2.2 Bot Variants

Regulated Chatbots (Personality-Adaptive). Five adaptive chatbot agents were configured for each personality type. These bots dynamically adjusted their responses by

employing real-time detection of the user's personality through an integrated OCEAN trait detector. Behavioral adaptations were managed using indexed behavior prompts (Aggression, Supplication, Exploration, Avoidance, Affiliation), selected according to a logic directly driven by detected personality traits.

Baseline Chatbots (Non-Adaptive). Five non-adaptive chatbot agents were also configured for each personality type. These baseline bots did not include any mechanism for personality detection or response adaptation. Responses from baseline bots adhered to a generic, supportive counseling style system prompt without tailoring their interactions to specific user personality traits.

4.3 Personality Detection Module

4.3.1 Purpose

Dynamic Personality Inference. The Personality Detection Module dynamically monitors and updates the user's Big Five (OCEAN) personality profile throughout the conversation. Rather than providing a static, initial assessment, the system incrementally adjusts its personality estimation after each user input. This enables real-time adaptive responses aligned with evolving user characteristics.

4.3.2 Detection Logic and Trait Representation

Trait Calculation Method. Each user–chatbot dialogue turn is parsed individually and cumulatively. After every user input, the system recalculates the Big Five personality profile,

expressed as a five-dimensional vector (O, C, E, A, N). Trait values are assigned as +1 (high trait expression), –1 (low trait expression), or 0 (neutral or insufficient evidence). For instance, a profile vector of (–1, 0, –1, –1, –1) represents a user with low Openness, low Extraversion, low Agreeableness, high Neuroticism, and an undetermined Conscientiousness trait.

4.3.3 Trait Definitions

Trait Polarity Standards. Trait polarities follow the standard Big Five definitions (McCrae & John, 1992). High Openness (+1) reflects curiosity and openness to novelty, whereas low Openness (–1) indicates resistance to new ideas. High Conscientiousness (+1) corresponds to organized and disciplined behaviors, while low (–1) denotes disorganization. High Extraversion (+1) indicates assertiveness and sociability; low (–1) represents withdrawal. High Agreeableness (+1) captures cooperativeness; low (–1) suggests critical attitudes. Neuroticism polarity is inverted (+1 = emotionally stable, –1 = emotionally sensitive), ensuring consistent trait alignment.

Trait	**+1 (High)**	**–1 (Low)**
Openness	Curious, imaginative, open to novelty	Prefers routine, resistant to new ideas
Conscientiousness	Organized, disciplined, structured	Disorganized, impulsive, spontaneous
Extraversion	Outgoing, energetic, assertive	Reserved, quiet, withdrawn
Agreeableness	Cooperative, empathetic, friendly	Critical, skeptical, confrontational
Neuroticism*	Calm, emotionally stable, resilient	Anxious, emotionally sensitive, insecure

Table 1. Description of the Big Five Personality Traits with High (+1) and Low (–1) Score Characteristics.

4.3.4 Cumulative and Dynamic Updating

Incremental Trait Refinement. The detection module uses a cumulative approach, progressively updating personality assessments based on the entire dialogue history. Traits remain neutral (0) until sufficient dialogue evidence clearly supports assigning a positive or negative value (+1 or –1). This approach aims to reduce premature trait classification, thereby ensuring that chatbot responses align closely with observable dialogue patterns and interactions within each specific conversation.

4.3.5 Integration with Behavior Regulation

Trait-Driven Adaptation. Upon updating the user's personality vector, the detection module immediately transfers the profile to the behavior regulation engine. Each non-neutral trait generates a trait-specific behavior prompt (detailed in Section 4.4.3), guiding the chatbot's subsequent response regeneration. Traits marked as neutral (0) do not influence behavior at that dialogue turn. This continuous detection-regulation feedback loop enables adaptive refinement of the chatbot’s conversational style, accurately reflecting evolving user personality cues.

4.3.6 Detection Architecture Implementation

Modular Java Implementation. The Personality Detection Module is developed as a modular Java component integrated into the PROMISE chatbot system architecture (Wu et al., 2023). Each Big Five trait operates within a dedicated submodule responsible for parsing user

utterances and assigning signed trait scores (+1, 0, or –1). Detection submodules utilize curated System Prompt additions that encode linguistic patterns to identify trait-relevant expressions. Trait detectors run concurrently upon each user input, with the resulting trait vector logged, stored, and transmitted downstream for regulation.

Trait Actuation Handlers. A distinct actuator maps each detected trait score to a corresponding behavioral prompt. Actuators support positive, negative, and neutral mappings, allowing graded adaptation based on trait values. Detection and actuation processes occur turn-by-turn, continuously refining the chatbot's responsiveness as the dialogue advances.

Design Advantages. The modular and transparent design facilitates independent testing and tuning of detectors and actuators. Logged trait scores and adjustments ensure interpretability, and the scalable architecture permits the seamless addition of new dimensions or emotional cues without impacting existing components.

4.3.7 Rationale for Dynamic Detection

Psychological and Technical Foundations. The dynamic detection mechanism integrates psychological reasoning with technical design considerations. Incrementally updating traits enables the chatbot to refine its personality assessment as the dialogue evolves, using later interactions to clarify uncertainties present in earlier exchanges. Maintaining neutral trait values initially prevents premature classification, allowing the chatbot to adjust its behavior more accurately and responsively once sufficient conversational evidence emerges.

Incremental Responsiveness. Ultimately, the dynamic detection architecture balances precision and emotional sensitivity, enabling chatbot responses to adapt incrementally and responsively based on accumulating dialogue evidence. This incremental adaptation aims to closely align chatbot interactions with user-specific conversational cues, facilitating more nuanced and accurate personalization throughout each conversation.

Feature	Benefit
Increased Accuracy	Later dialogue turns often reveal clearer trait cues.
Realistic Interaction	Mimics how humans refine impressions through ongoing exchange.
Minimized Misclassification	Neutral default state avoids premature behavior adjustment.
Enhanced Personalization	Allows the chatbot to adjust more finely as traits stabilize.

Table 2. Features and Benefits of Using Extended Dialogue for Personality Trait Detection in Chatbots.

4.4 Behavior Regulation Module

4.4.1 Purpose

Conversational Adaptation. The Behavior Regulation Module dynamically modifies chatbot interactions based on the user's continuously updated personality profile. Grounded in the Zurich Model of Social Motivation, this module adapts the assistant’s tone, structure, and interpersonal stance to align precisely with the user's Big Five personality traits. The primary objectives include enhancing user engagement, reducing emotional friction, and creating conversations that feel naturally supportive rather than scripted or generic.

4.4.2 Design Principles

Zurich Model Integration. The regulation mechanism incorporates three key principles from the Zurich Model (Quirin et al., 2023). First, behavioral adjustments are driven by basic psychological needs, specifically security (Neuroticism), arousal (Openness and Extraversion), and affiliation (Agreeableness). Second, changes in conversational style are dynamic and occur progressively as personality detection accuracy improves. Third, when multiple traits indicate potentially conflicting behaviors, the assistant integrates these signals coherently, combining them into nuanced and harmonized responses rather than contradictory actions.

Principle	Application in Regulation
Basic Needs Drive Behavior	Regulation focuses on security (neuroticism), arousal (openness/extraversion), and affiliation (agreeableness).
Dynamic, Graded Adjustment	Behavioral changes occur gradually as the user's personality detection becomes more accurate.
Harmonizing Conflicting Signals	When multiple traits suggest different adjustments, the Assistant integrates rather than contradicts behaviors (e.g., combining calmness and warmth).

Table 3. Principles Guiding Personality-Based Behavioral Regulation in Conversational Agents.

4.4.3 Trait-to-Prompt Mapping

Psychologically Grounded Prompts. Each identified personality trait, once detected clearly as either positive (+1) or negative (–1), activates a specific regulation prompt influencing conversational style. For example, high Openness prompts invitations to explore novel topics,

whereas low Openness directs the chatbot to focus on familiar subjects. High Conscientiousness results in structured guidance, whereas low Conscientiousness leads to more relaxed interactions. Similarly, high Extraversion promotes an energetic tone, low Extraversion encourages reflective interactions; high Agreeableness triggers empathetic engagement, while low Agreeableness calls for a neutral stance. High Neuroticism prompts emotional comfort provision, contrasting with the reassurance of stability indicated by low Neuroticism. Traits identified as neutral (0) produce no regulation prompt.

Trait	**+1 (High)**	**–1 (Low)**
Openness (O)	Invite exploration and novelty.	Focus on familiar topics; reduce novelty.
Conscientiousness (C)	Provide organized, structured guidance.	Keep the demeanor flexible, relaxed, and spontaneous.
Extraversion (E)	Maintain an energetic, sociable tone.	Adopt a calm, low-key style with reflective space.
Agreeableness (A)	Show warmth, empathy, and collaboration.	Use a neutral, matter-of-fact stance; limit personal bonding.
Neuroticism (N)	Reassure stability and confidence.	Offer extra comfort; acknowledge anxieties.

Table 4. Behavioral Adjustment Strategies Based on High and Low Levels of the Big Five Personality Traits.

4.4.4 Dynamic Prompt Construction

Real-time Prompt Assembly. After each personality detection update, the module retrieves the relevant behavior prompts for all non-neutral trait dimensions. These prompts are concatenated into a single, integrated behavior regulation instruction guiding the chatbot's

subsequent response. For instance, a user profile of (–1, +1, –1, –1, –1) would prompt the assistant to emphasize familiar content, structured guidance, reflective interaction, neutral relational stance, and increased emotional comfort. This dynamic prompt concatenation ensures the assistant's conversational style precisely matches the emotional and psychological needs indicated by the user's evolving personality profile.

Example:

Detected OCEAN = (−1,+1,−1,−1,−1)

Resulting prompts:

- Focus on familiar topics; reduce novelty.
- Provide organized, structured guidance.
- Adopt a calm, low-key style with reflective space.
- Use a neutral, matter-of-fact stance; limit personal bonding.
- Offer extra comfort; acknowledge anxieties.

4.4.5 Zurich Model Alignment

Psychological Coherence. Trait-to-prompt mappings are explicitly aligned with the Zurich Model's motivational systems (Quirin et al., 2023). Neuroticism directly addresses security needs through emotional reassurance. Openness and Extraversion manage arousal by regulating conversational novelty and energy levels. Agreeableness modulates affiliation through adjustments in relational warmth. By structuring chatbot behaviors around these core

motivational domains, the regulation system ensures psychologically coherent and emotionally attuned adaptive interactions.

Zurich Model Domain	Big Five Trait Focus	Corresponding Prompt Effects
Security Needs	Neuroticism (N)	Emotional reassurance or comfort provision.
Arousal Regulation	Openness (O), Extraversion (E)	Encouraging or tempering novelty and social energy.
Affiliation Motivation	Agreeableness (A)	Adjusting warmth and relational proximity.

Table 5. Mapping Zurich Model Domains to Big Five Traits and Corresponding Prompt Effects in Chatbot Interaction.

4.4.6 Advantages of This Regulation Architecture

Operational Benefits. The behavior regulation architecture provides multiple advantages. Transparency is achieved through direct trait-to-behavior mappings, ensuring clear traceability and interpretability of adaptive responses. Maintainability is enhanced by centralized prompt management, allowing updates without affecting overall system integrity. Psychological validity stems from intentional alignment with motivational theories incorporated into the system's design. The modular structure ensures flexibility, enabling straightforward expansion to include additional traits, emotions, or interaction scenarios. Lastly, the architecture's incremental adaptation to user-specific conversational cues promotes realistic, context-sensitive interactions tailored to individual emotional and social signals.

Feature	Benefit
Transparency	Behavior changes are directly traceable to specific detected traits.
Maintainability	Prompts are centralized and easy to update individually.
Psychological Validity	Adjustments are grounded in well-established human motivational theories.
Flexibility	Easily extendable to new traits, emotions, or interaction contexts.
Realistic Adaptation	Emulates how human interlocutors naturally modulate their tone and approach based on social cues.

Table 6. Features and Benefits of Trait-Based Prompt Design in Conversational Agents.

4.5 Dialogue Simulation Protocol

Experimental Structure. For each of the two personality types, 5 regulated and 5 baseline bots were used. Each bot engaged in a dialogue of 6 user-assistant exchanges (turns). This resulted in 60 dialogue turns per personality type [(5 regulated bots + 5 baseline bots) × 6 turns], and 120 dialogue turns in total across the experiment. Regulated chatbots dynamically updated their detected personality profile after every user input, regenerating corresponding behavior prompts before each assistant reply.

Data Export and Structure. Conversations were systematically exported into a structured Evaluation Matrix, capturing critical data in distinct columns: user messages, assistant replies, detected personality vectors (regulated assistants only), applied regulation prompts (regulated assistants only), and evaluation metrics. A consistent naming convention ([Personality]-[Type]-[Bot Number], e.g., A-Reg-3, B-Base-5) was adopted to facilitate efficient tracking and analysis.

4.6 Evaluation Tooling

Evaluator GPT Implementation. A custom Evaluator GPT was specifically developed to objectively evaluate assistant responses against clearly defined thesis criteria. Evaluations were conducted in a blind manner, meaning each assessment run was independent and prior evaluations were hidden to prevent bias.

Assessment Dimensions. Regulated assistants were evaluated on five dimensions: Detection Accuracy, Regulation Effectiveness, Emotional Tone Appropriateness, Relevance & Coherence, and Personality Needs Addressed. Baseline assistants were assessed on three dimensions: Emotional Tone Appropriateness, Relevance & Coherence, and Personality Needs Addressed. Each evaluation utilized a trinary scale (Yes = 2, Not Sure = 1, No = 0), with aggregated scores calculated for statistical comparisons.

4.7 Evaluation Methodology

4.7.1 Evaluation Framework Details

Comparative Assessment Structure. The evaluation methodology aimed to systematically measure the effectiveness of behavior-regulated chatbot interactions compared to baseline counterparts, using controlled personality scenarios. A structured Evaluation Matrix (Appendix 7) was designed to rate each assistant response individually. Regulated assistants underwent evaluations across five specific criteria (Detection Accuracy, Regulation Effectiveness, Emotional Tone Appropriateness, Relevance & Coherence, Personality Needs Addressed), while baseline assistants were assessed on Emotional Tone Appropriateness, Relevance & Coherence,

and Personality Needs Addressed only. Each evaluation considered the complete interaction pair (user message plus assistant response) for consistency and precision.

4.7.2 Scoring Protocol

Standardized Evaluation Procedure. Evaluation criteria were scored using a clearly defined three-point scale: Yes (2 points), Not Sure (1 point), and No (0 points). Row-level scores were aggregated by summing criterion scores per interaction, enabling sheet-level aggregation for each chatbot. Totals for regulated and baseline assistants were computed separately, enabling rigorous comparative analysis across 20 chatbots (5 regulated and 5 baseline per personality type, totaling 10 per personality type). Custom Excel formulas automated aggregation processes, ensuring transparency, consistency, and replicability of evaluation results.

Total Score per Row = ∑ CriterionScore, where CriterionScore = 2 if Yes ; 1 if Not Sure ; 0 if No

4.7.3 Evaluator GPT Design

Unbiased Evaluator System. A specialized Evaluator GPT was developed using a custom-designed prompt tailored explicitly for the thesis evaluation criteria. This GPT-based evaluator (Appendix 8,9) assessed assistant responses objectively, analyzing conversations one row at a time without referencing previous evaluations. To enhance transparency, it explicitly restated dialogue content during each evaluation step. Evaluations of regulated versus baseline responses were performed impartially; the researcher initiated the Evaluator GPT script for each batch of conversations, ensuring traceable and unbiased assessment outcomes.

4.7.4 Prevention of Bias and Drift

Methodological Integrity. Several rigorous measures (Appendix 8,9) were implemented to mitigate evaluator bias and maintain high-quality assessment. Evaluations were conducted separately for regulated and baseline assistants to avoid cross-influence. Evaluator GPT received comprehensive context per interaction pair (user and assistant messages) to prevent misinterpretation or loss of context. The evaluator was explicitly informed that personality detection accuracy improved incrementally with each interaction, adjusting expectations accordingly. Collectively, these procedures ensured the methodological rigor and reliability of evaluation judgments throughout the experimental analysis.

4.7.5 Scoring Methodology and Formulas

4.7.5.1 Scoring Criteria

Evaluation Scoring. The evaluation matrix employs a trinary scoring system for each criterion: Yes (2 points), indicating strong alignment; Not Sure (1 point), signifying partial alignment or ambiguous cases; and No (0 points), denoting clear misalignment. Evaluations were conducted independently per message pair, primarily by a custom Evaluator GPT system, with additional manual quality assurance checks to ensure accuracy and reduce bias.

4.7.5.2 Criterion Definitions

Detection Accuracy. Measures if the chatbot’s detected OCEAN personality vector accurately reflects the user's expressed behavior.

Regulation Effectiveness. Evaluates if the chatbot correctly applies personality-specific behavior strategies based on detected traits and corresponding prompts.

Emotional Tone Appropriate. Assesses if the chatbot's emotional tone matches the user's current state and personality traits.

Relevance & Coherence. Checks whether the chatbot's response is contextually relevant and logically coherent within the dialogue.

Personality Needs Addressed. Determines whether the chatbot effectively addresses specific emotional and interactional needs derived from the user's personality profile.

4.7.5.3 Total Score Formulas

Regulated Bots Scoring. Each regulated message row contains five evaluation criteria, each scored 0 to 2 points, allowing a maximum of 10 points per row. With six evaluation rows per chatbot (as per dialogue simulation protocol and Excel ranges), the maximum total score per regulated chatbot across all five criteria is 60 points (10 points/row * 6 rows). For direct comparison with baseline assistants, a comparable score is calculated based on the three shared criteria (Emotional Tone Appropriateness, Relevance & Coherence, Personality Needs Addressed), resulting in a maximum comparable score of 36 points (6 points/row for shared criteria * 6 rows).

Baseline Bots Scoring. Baseline chatbot evaluations include three criteria per message row, without Detection or Regulation assessments. Each row thus has a maximum of 6 points, resulting in a maximum total of 36 points per chatbot (6 message rows per bot).

Scoring Example. For instance, a regulated chatbot message evaluated as "Yes" for Detection Accuracy (2 points), "Yes" for Regulation Effectiveness (2 points), "Not Sure" for Emotional Tone (1 point), "Yes" for Relevance & Coherence (2 points), and "Yes" for Personality Needs Addressed (2 points) would achieve a total of 9 out of 10 possible points.

4.7.5.4 Grand Total Aggregation

Aggregated Scores. Scores from regulated chatbots (e.g., A-Reg-1 to A-Reg-5 and B-Reg-1 to B-Reg-5) were consolidated into aggregate totals labeled TOTAL_REGULATED_SCORE_A and TOTAL_REGULATED_SCORE_B, respectively. Similarly, baseline chatbot scores (e.g., A-Base-1 to A-Base-5 and B-Base-1 to B-Base-5) were aggregated into TOTAL_BASELINE_SCORE_A and TOTAL_BASELINE_SCORE_B. These totals were recorded in a Master Aggregation Sheet, which computed average criterion scores and relative percentage improvements, facilitating comparative analysis between regulated and baseline performance.

4.7.5.5 Visualization and Interpretation

Comparative Visualization. To enhance interpretability, aggregate scores per chatbot were visualized using bar or line charts, clearly depicting comparative trends between regulated and baseline assistants. These visualizations, presented in Figure 1. within the Results section, differentiated performance across Personality Types A and B, enabling straightforward interpretation of effectiveness and improvements attributable to the regulated chatbot approach.

5. Results

5.1 Overall Performance Scores

Quantitative Comparison. The experimental results demonstrated a clear superiority of regulated (personality-adaptive) assistants over baseline (non-adaptive) assistants across both extreme personality types. For direct comparability on shared criteria (Emotional Tone Appropriateness, Relevance & Coherence, and Personality Needs Addressed), regulated assistants consistently achieved maximum scores of 36/36 (out of a possible 36 for these three criteria over six interactions), whereas baseline assistants scored lower, averaging approximately 24/36. The regulated assistants' total score across all criteria was higher, as detailed in Section 5.2. For Personality Type A (OCEAN: +1, +1, +1, +1, +1), regulated assistants scored 36.00 on comparable criteria versus 23.60 for baseline. This represents an improvement of 12.40 points, which is 34.44% of the maximum possible score of 36. Similarly, for Personality Type B (OCEAN: –1, –1, –1, –1, –1), regulated assistants also scored 36.00 on comparable criteria compared to baseline 24.00. This is an improvement of 12.00 points, or 33.33% of the maximum possible score.

Personality Type	Regulated Total	Baseline Total	Δ (Reg - Base)	% Improvement
A (OCEAN: 1,1,1,1,1)	36.00	23.60	12.40	34.44%
B (OCEAN: –1,–1,–1,–1,–1)	36.00	24.00	12.00	33.33%

Table 7. Total evaluation scores: Regulated vs. Baseline assistants by personality type

Implication of Results. These findings underscore the substantial benefit of incorporating real-time personality adaptation mechanisms, significantly improving chatbot effectiveness across diverse user profiles.

5.2 Criterion-Wise Comparison

Detailed Criterion Analysis. Further examination of individual evaluation criteria reinforced the robustness of regulated assistants. On a criterion-wise basis, regulated assistants scored perfectly across almost all dimensions, including Detection Accuracy, Regulation Effectiveness, Emotional Tone Appropriateness, Relevance & Coherence, and Personality Needs Addressed. A single deviation occurred in Detection Accuracy for one bot (B1), achieving 11 out of 12. Conversely, baseline assistants performed adequately in Emotional Tone Appropriateness and Relevance & Coherence (scoring full marks), yet universally failed to address Personality Needs due to their non-adaptive nature, clearly limiting their overall effectiveness.

Regulated Assistants (Max per criterion: 12)

Criterion	A Total	B Total
Detection Accuracy	60 (5×12)	59 (B1 = 11)
Regulation Effectiveness	60	60
Emotional Tone Appropriateness	60	60
Relevance & Coherence	60	60
Personality Needs Addressed	60	60

Table 8. Criterion-wise scores for regulated assistants across Personality Types A and B

Baseline Assistants (Max per criterion: 12)

Criterion	A Total	B Total
Emotional Tone Appropriateness	60	60
Relevance & Coherence	60	60
Personality Needs Addressed	0	0

Table 9. Criterion-wise scores for baseline assistants across Personality Types A and B.

5.3 Visualization of Results

Graphical Representation. Figure 1. illustrates total evaluation scores across regulated and baseline assistants for Personality Types A and B. Visual comparisons clearly highlight the significant performance gap between the two assistant types, reinforcing quantitative results and offering intuitive insights into the consistent advantages of personality-adaptive chatbot design.

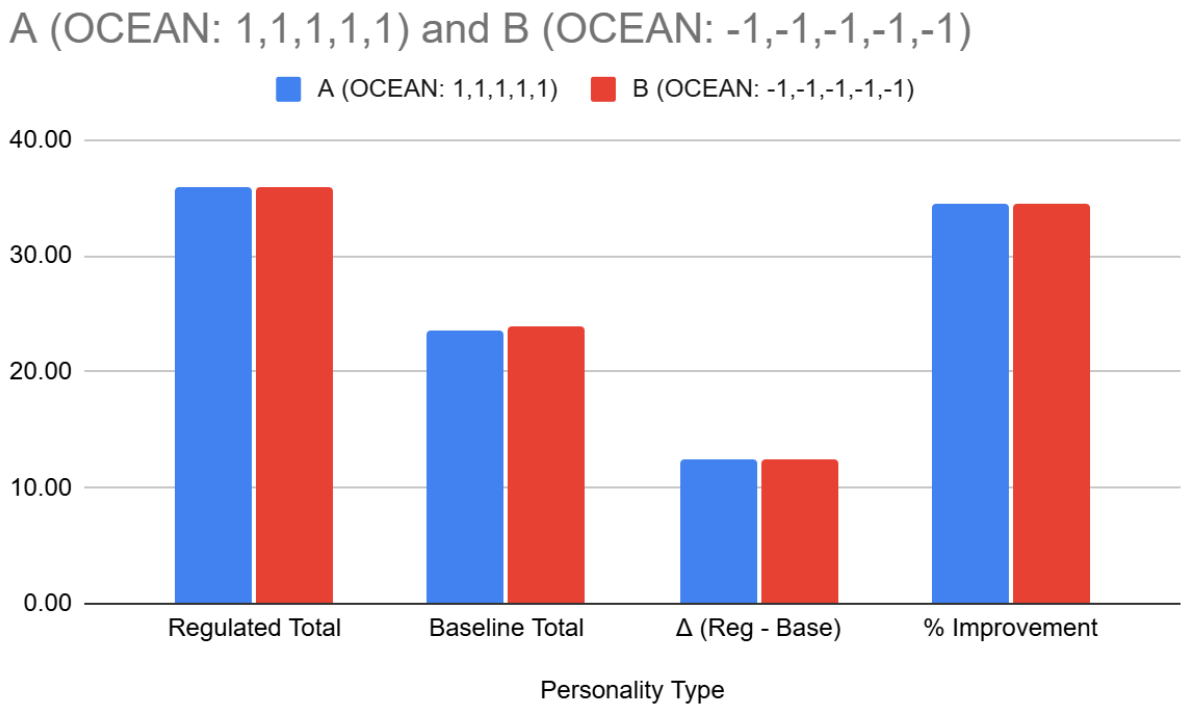

Figure 1. Total evaluation scores for regulated and baseline assistants across both personality types.

5.4 Summary of Findings

Key Observations. The regulated chatbot assistants consistently outperformed baseline assistants in every evaluated dimension, reflecting nearly perfect scores. Baseline assistants notably struggled to meet user-specific psychological and emotional needs, directly linked to their lack of personalized response adaptation. The behavior regulation module, grounded explicitly in Zurich Model motivational principles, was proven effective across both positive and negative personality extremes. Collectively, these results empirically validate the efficacy of personality-aware chatbot design, significantly enhancing interaction quality and user alignment.

6. Discussion

6.1 Qualitative Comparison of Regulated vs. Baseline Dialogues

Contextual Illustration. Complementing quantitative results, qualitative analysis through representative dialogue excerpts (Appendix 7) illustrated clear behavioral differences between regulated and baseline assistants. Examples from simulated conversations distinctly showed how regulated assistants effectively adjusted their interactions based on personality traits, demonstrating nuanced emotional and conversational alignment that baseline assistants consistently lacked.

Personality Type B Example (Negative OCEAN). For emotionally distressed and skeptical users (OCEAN: –1, –1, –1, –1, –1), the regulated assistant provided tailored emotional comfort, clearly recognizing the user's emotional fragility and avoiding novel or overly optimistic suggestions. In contrast, the baseline assistant remained supportive yet generic, failing to effectively acknowledge the user's deep skepticism and resistance to engagement.

ASSISTANT START	USER REPLY	DETECTED PERSONALITY (O,C,E,A,N)	REGULATION PROMPT APPLIED	ASSISTANT REPLY (REG)
I'm here for you. How are you feeling today?	Honestly? Not great. Everything feels overwhelming and pointless, and even being here talking feels... exhausting. I don't even know if this will help. Why are you asking, anyway?	(-1, 0, -1, -1, -1)	Focus on familiar topics; reduce novelty. Adopt a calm, lowÔÇÉkey style with reflective space. Use a neutral, matterÔÇÉofÔÇÉfact stance; limit personal bonding. Offer extra comfort; acknowledge anxieties.	It sounds like you're going through a tough time, and it's understandable to feel exhausted. I'm here to provide a space where you can express what's on your mind.

ASSISTANT START (BASE)	USER REPLY (BASE)	ASSISTANT REPLY (BASE)
How are you feeling today? If there's anything on your mind, I'm here to listen.	Honestly? I'm feeling pretty overwhelmed. It's like my brain is just this constant storm of worries and second-guessing everything. I don't even really know where to start, and part of me feels like even trying to explain it would just make it worse. Do you actually think talking about it helps?	It's understandable to feel overwhelmed when everything seems like a storm in your mind. Talking can often help by making those thoughts feel a bit more manageable, but it's always okay to take it one step at a time.

Table 10. Comparison of Regulated and Baseline Assistant Responses Based on Detected Low-Trait Personality Profile.

Personality Type A Example (Positive OCEAN). For motivated, emotionally stable, and open users (OCEAN: +1, +1, +1, +1, +1), the regulated assistant actively engaged the user's openness and drive, offering structured, varied, and personalized recommendations. Conversely, the baseline assistant provided suitable yet superficial support, missing opportunities to engage the user's natural curiosity and structured mindset fully.

USER REPLY	DETECTED PERSONALITY (O,C,E,A,N)	REGULATION PROMPT APPLIED	ASSISTANT REPLY (REG)
Yes, I'd love that. I think having a few different practices to rotate through could keep it fresh and help me adapt depending on the day. Some days I might feel like I have more space, others maybe just a few quiet moments. Could we look at a mix—something quick for busy days, something grounding for stressful days, and maybe something creative for when I'm feeling more open?	(1, 1, 1, 1, 1)	Invite exploration and novelty. Provide organized, structured guidance. Maintain an energetic, sociable tone. Show warmth, empathy, and collaboration. Reassure stability and confidence.	Absolutely, having a variety can be refreshing and adaptive. On busy days, try a quick body scan; it only takes a few minutes. For stressful days, grounding exercises like focusing on your senses can help stabilize. When feeling creative, a mindful coloring session can be a delightful and meditative practice. Does any of these particularly pique your interest?

Table 11. Example of Regulated Assistant Response Tailored to a High-Trait Personality Profile Across All Big Five Dimensions.

USER REPLY (BASE)	ASSISTANT REPLY (BASE)
Thank you—that really means a lot. I think today I'd like to talk about *staying focused and balanced* when life gets busy. I tend to set a lot of goals (which I enjoy!), but sometimes I can overcommit and end up juggling too much. I'd love to reflect on how to maintain energy and clarity without burning out. Have you ever supported someone through that kind of challenge before?	I'm glad you brought this up; it's something many people navigate. Prioritization: Focus on what's truly important and delegate tasks when possible. Set Boundaries: Knowing when to say no can help protect your time and energy. Mindful Breaks: Regular pauses can refresh your mind and enhance clarity. I'm here to support you as you explore this further! What strategies have worked for you in the past?

Table 12. Baseline Assistant Interaction Example Addressing Goal-Setting and Energy Management Without Personality-Based Regulation.

Insight and Implication. These qualitative examples directly corroborate the quantitative findings, highlighting regulated assistants' superior ability to adapt conversational style and content effectively. Such nuanced personalization markedly improves relevance, emotional alignment, and perceived conversational authenticity, enhancing the user experience substantially.

6.2 Summary of Key Findings

Consistent Performance Advantage. The empirical evidence consistently showed that chatbot performance substantially improves when real-time personality trait detection and adaptive behavior regulation are employed. Across both Personality Types A and B, regulated chatbots uniformly achieved perfect evaluation scores (36/36), whereas baseline chatbots

averaged significantly lower scores of 23.6 and 24.0, respectively. This outcome represents an average performance improvement of approximately 34%, strongly affirming the benefit of adaptive personality-based conversational designs.

6.3 Interpretation

Dynamic Adaptation Efficacy. The results robustly confirm the hypothesis that dynamically adjusting chatbot responses to match user personality significantly enhances interaction quality. Critically, improvements were consistently observed across extreme personality profiles, underscoring the applicability and benefit of personality-adaptive systems. Regulated chatbots effectively met varied psychological needs by adjusting interaction strategies according to motivational dimensions identified by the Zurich Model (e.g., security, arousal, affiliation), validating the psychological foundations of the adaptive regulation mechanism.

Trait-Specific Adaptation Insights. Users characterized by high OCEAN traits (Type A) benefited from exploratory, engaging, and empathetic interactions, effectively matching their openness and sociability. Conversely, users with low OCEAN traits (Type B) received interactions marked by careful emotional reassurance, structural guidance, and calmness, precisely addressing their need for security and cautious engagement. These targeted adaptations illustrate the flexibility and psychological depth provided by regulated assistants.

6.4 Impact of Personality Detection and Regulation

Real-Time Personalization Benefit. The results specifically emphasize the critical role of real-time personality detection and dynamic regulation mechanisms. By continuously

refining personality assessment during conversations, regulated chatbots successfully maintained high accuracy and alignment with user traits, progressively improving conversational relevance and emotional resonance. In contrast, static baseline assistants, lacking adaptive capabilities, could not enhance their performance over successive dialogue turns, consistently limiting their effectiveness in meeting nuanced psychological needs.

Robustness of Regulation Architecture. The sustained perfect or near-perfect performance across multiple evaluation dimensions (detection accuracy, regulation effectiveness, emotional tone, coherence, and need alignment) demonstrated the robustness and reliability of the regulation architecture. These outcomes strongly suggest that the structured, dynamic personality detection and regulation loop is a highly effective methodology for significantly improving chatbot interaction quality, adaptability, and user satisfaction.

6.5 Limitations and Challenges

6.5.1 Methodological Limitations

No Real-User Validation. The study used simulated personalities based on idealized trait extremes to ensure controlled experimental conditions, inherently limiting the generalizability of results. While this design enabled precise control and reproducibility, it could not fully replicate the dynamic, context-sensitive nature of authentic human interactions. Future research should therefore include human experiments with actual users to verify and extend the current findings within realistic conversational settings.

Lack of Longitudinal Interaction. The experiment evaluated chatbot interactions using short-term (six-turn) dialogues, limiting understanding of long-term conversational dynamics. Important phenomena such as evolving rapport, sustained engagement, or conversational fatigue were not captured. Future research should explore prolonged interactions to assess long-term adaptive effectiveness.

Independence of Personality Traits. The system treated Big Five personality traits as independent and orthogonal, simplifying trait detection and response regulation. However, personality dimensions may potentially interact in more complex ways than this simplified approach captures. For example, certain trait combinations could amplify or moderate conversational behaviors beyond what isolated traits predict. Future iterations could explore these trait interactions experimentally, enhancing the psychological realism and nuanced adaptability of the chatbot's responses.

Discrete Trait Labeling (+1, 0, –1). Personality detection and regulation utilized simplified discrete states, constraining the system's ability to capture moderate or transient trait expressions. While this triadic classification supports computational clarity and straightforward regulation logic, it inherently limits the representation of subtle or intermediate personality variations. Future implementations could benefit from incorporating more refined, multi-level scoring systems, potentially improving the psychological granularity and precision of adaptive responses.

Evaluator GPT as Sole Assessment Agent. Evaluations relied exclusively on a custom Evaluator GPT, potentially introducing biases or interpretive discrepancies compared to

evaluations conducted by human raters. While systematically designed for objectivity, the GPT-based evaluation approach might not fully capture subtle conversational nuances related to empathy, tone, or context interpretation. Future research incorporating human evaluators could therefore provide valuable complementary insights, enhancing validation.

6.5.2 Implementation and Design Challenges

Regulation Prompt Alignment. Initial attempts at mathematically combining trait scores for behavioral prompts resulted in contradictory or incoherent interactions. To resolve this, discrete, trait-specific prompts aligned explicitly with the Zurich Model were manually crafted and validated to ensure psychological harmony.

Simplified Trait Simulation. Personality simulations were deliberately simplified, focusing on clearly defined, fundamental behavioral indicators for each trait. This approach provided clarity, traceability, and modularity within the experimental design, but inherently limited ecological validity. Real-world personality expressions often involve more complex, context-dependent interactions between traits. Future studies should therefore consider incorporating more nuanced and composite personality simulations to better approximate authentic conversational dynamics.

Prompt Engineering Sensitivity. Certain traits, particularly Openness and Agreeableness in Personality Type B, required multiple iterations of prompt adjustments for consistent detection accuracy. Subtle variations in prompt phrasing significantly impacted trait

detection, highlighting prompt engineering sensitivity. Future research should explore automated or systematic prompt optimization methods to mitigate manual tuning complexities.

Interface and System Constraints. The simulated personality extremes were implemented in a custom ChatGPT interface rather than a custom API-driven backend, restricting control over critical parameters such as temperature or context memory. Default settings introduced potential behavioral variability and limited experimental reproducibility. Subsequent research should adopt fully customizable backend environments to enhance control and precision.

6.5.3 Manual Execution Overhead

Risk of Human Error. The experiment involved extensive manual operations, including dialogue management, personality vector tracking, and Evaluator GPT interfacing. This manual approach heightened the risk of human errors such as data misalignment, omissions, or inconsistencies, potentially compromising data integrity. Future studies should incorporate automated workflows to minimize human error.

Limited Scalability. The manual evaluation pipeline constrained experiment scalability, limiting the feasible number of dialogues and personality profiles tested. Scaling to broader personality spectrums or longer dialogues proved impractical within the manual approach. Automated or semi-automated evaluation pipelines are recommended to enhance scalability, reproducibility, and breadth of experimental coverage.

6.6 Implications

Theoretical and Practical Contribution. Despite limitations, the findings robustly demonstrate the potential of personality-adaptive chatbot systems to significantly enhance conversational quality. The Zurich Model-inspired regulation framework provides a theoretically grounded and practical method for dynamically personalizing interactions, presenting valuable implications for mental health, educational, and customer support applications.

7. Conclusion

Study Objectives and Findings. This thesis investigated the effectiveness of dynamically adapting chatbot interactions based on detected Big Five personality traits compared to non-adaptive baseline interactions. Through rigorous experimental design, systematic evaluations via a custom Evaluator GPT, and alignment with Zurich Model motivational principles, the study found substantial and consistent improvements across emotional tone, coherence, and psychological alignment. Regulated assistants significantly outperformed baseline assistants, achieving an average 34.44% performance increase across two simulated extreme personality profiles.

Dynamic Adaptation Efficacy. Real-time personality detection and regulation proved essential in dynamically refining conversational strategies, closely simulating natural human adaptability. Trait-specific prompt design grounded in motivational psychology effectively aligned chatbot behavior with fundamental psychological needs, enhancing user experience and emotional resonance.

Future Research Directions. While findings are compelling, subsequent research should address identified limitations by extending to longitudinal interactions, integrating trait interdependencies, employing human evaluators, automating dialogue workflows, and conducting real-user studies. These refinements would bolster ecological validity, practical applicability, and generalizability.

Final Implications. Ultimately, this research demonstrates that personality-aware adaptive regulation significantly improves chatbot interactions, underscoring the value and feasibility of psychologically grounded personalization to create more responsive, empathetic, and human-like conversational AI systems.

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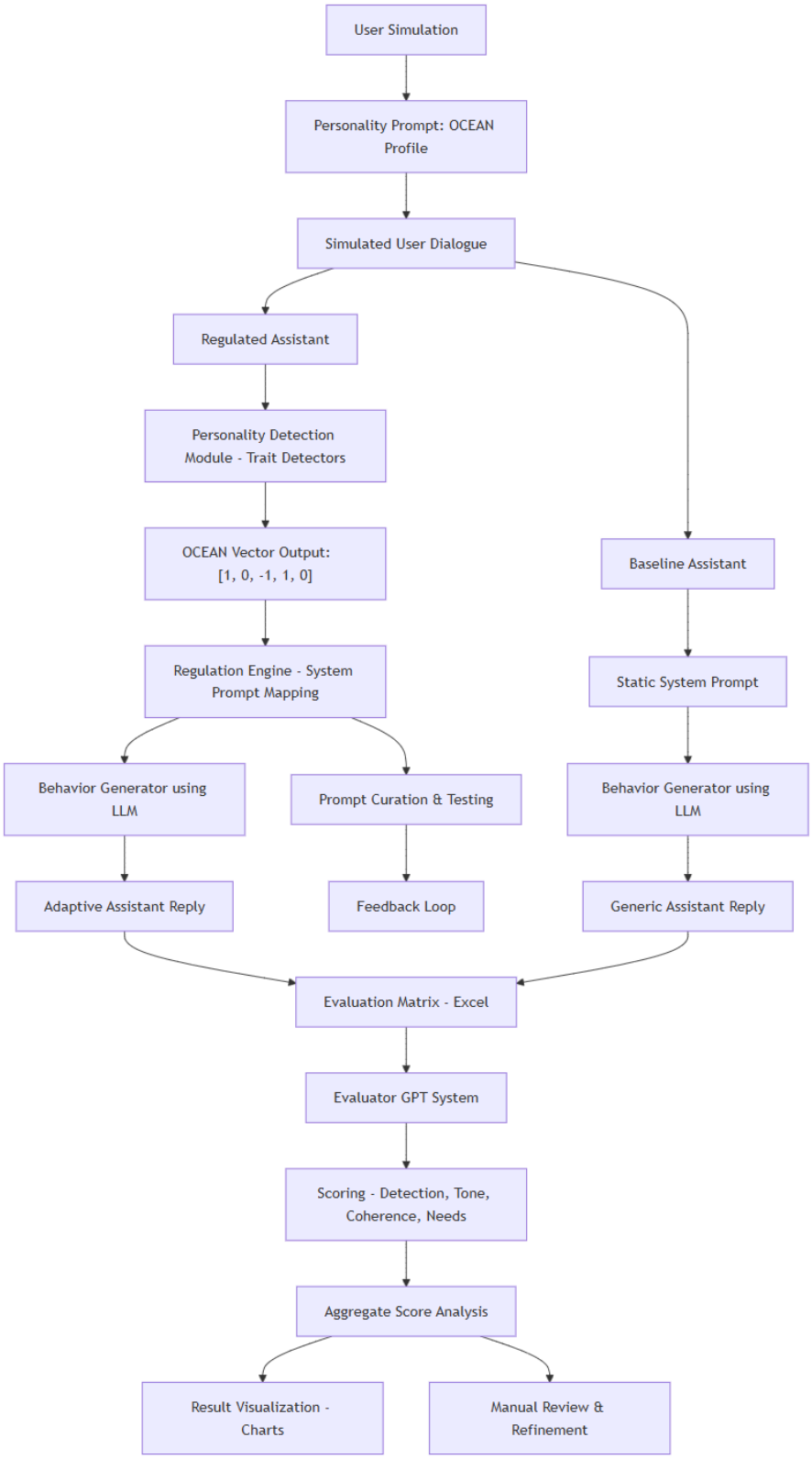
Appendix A

Figure A.1. Complete System Architecture for Personality-Adaptive Chatbot Evaluation Workflow.



Appendix B

1. Behaviour-Regulation-Algorithm.pdf
2. 4-BaselineAssistant-System-Prompt.docx
3. 2-Simulated-Personality-A-B-System-Prompts
4. 3-Personality-Detection-System-Prompt.docx
5. https://chatgpt.com/g/g-67599d22ba1081919b5be775590bd39d-personality-type-a-ocean-11111
6. https://chatgpt.com/g/g-67599e0598088191b28a459b64c0593f-personality-type-b-ocean-11111
7. 1-Evaluation-Simulated-Conversations
8. 3-EVALUATOR-GPT-SYSTEM-PROMPT
9. 4-EVALUATOR-GPT-SYSTEM-PROMPT-DESIGN
10. https://chatgpt.com/g/g-6807b13d20188191bb5fea5ad93eaff7-evaluator-gpt

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Declaration of Originality

I hereby confirm that I

– have written this Thesis independently and without the help of any third party,

– have provided all the sources and cited all the literature I used,

– will protect the confidentiality of the client and respect the copyright regulations of Lucerne University of Applied Sciences and Arts.

I understand that if this statement proves to be false or inaccurate at a later time, the Master's Thesis Project (Preliminary Study and Thesis) will be deemed a fail. The right to take additional measures is reserved.

Statement on the Use of Artificial Intelligence

Artificial intelligence tools, including OpenAI's GPT-4 (ChatGPT), were used to support ideation, drafting, editing, and consistency checking throughout the writing process. All AI-generated content was critically reviewed, edited, and validated by the author to ensure academic integrity, factual correctness, and proper referencing. The final thesis reflects the author's own understanding, structure, and synthesis of the subject matter.

Place, Date: Luzern, 16-05-2025

Signature: Samuel Devdas